Language processing demands predict biological brain age: MRI evidence from bilinguals, translators, and interpreters

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Background. Simultaneous interpreting is an extreme case of incremental sentence processing: comprehension, production, prediction, and language control must run in parallel under severe time pressure (Korenar et al., 2023). While bilingual experience reliably reshapes language control circuits (DeLuca et al., 2020), it remains unclear how sustained high demand online processing relates to neurobiological markers of brain ageing. We tested whether increasing levels of bilingual language processing expertise (bilinguals → translators → simultaneous interpreters) are associated with distinct structural brain age profiles.

Methods. We analysed high resolution T1 MRI from 103 right handed native Czech speakers (bilinguals n=42; translators n=35; interpreters n=26). T1 scans were processed with FastSurfer (DKT parcellation) to extract regional grey/white matter and CSF volumes and intracranial volume (ICV). Brain age was estimated using six published pretrained models (BrainAge, BrainAgeR, DeepBrainNet, Pyment, BRAID GM, BRAID WM) (e.g., Wood et al., 2024). For each model we computed brain age gap (BAG = predicted − chronological age) and applied 5 fold cross validated age bias correction. To link expertise to candidate processing circuits, we examined (i) group differences in age BAG slopes (Age × Group) and (ii) ROI wise associations between corrected BAG and ICV/age/sex adjusted regional volumes (FDR across ROIs).

Results. Across models, raw BAG showed a reliable Group × Model interaction: interpreters tended to exhibit more positive BAG than bilinguals in several models, with translators intermediate. After cross validated age bias correction, global group offsets largely attenuated. Crucially, within the BrainAge model, age BAG relationships differed by expertise (Fig. 1): raw BAG decreased with age in all groups, but slopes became progressively steeper from bilinguals (β= 0.34) to translators (β= 0.58) to interpreters (β= 0.73), and interpreters differed from bilinguals. For corrected BAG, bilinguals showed a slight positive age slope (β=0.22), translators were near flat, and interpreters showed a negative trend (β= 0.20), yielding a significant Age × Group interaction. ROI wise analyses further suggested distinct neuroanatomical signatures: in bilinguals, higher corrected BAG correlated with larger CSF/ventricular volumes (global ageing markers), whereas in interpreters, corrected BAG related to orbitofrontal and ventral diencephalic volumes (Fig. 2), regions implicated in value based updating and high level control that support rapid selection during concurrent comprehension/production.

Conclusion. Structural brain age modelling when combined with bias correction and regional analyses detects graded differences across bilingual expertise levels that align with the language processing demands of real time language use. Our results suggest that sustained engagement in high pressure incremental sentence processing (interpreting) is associated with a more focal prefrontal/subcortical structure brain age coupling than routine bilingual use, motivating brain age approaches as a complementary marker for experience dependent adaptation in the language control system.

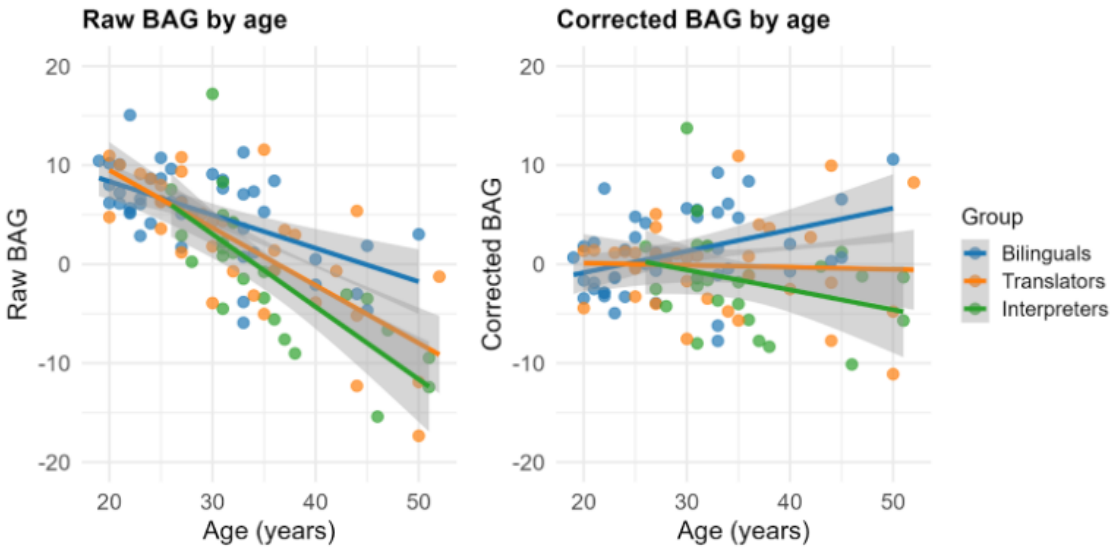

Figure 1. Relationship between age and brain age gap (BAG) by language-experience group. Left: raw BAG; right: cross-validated age-corrected BAG. Points represent individual participants; lines show group-wise linear fits with 95% confidence intervals.

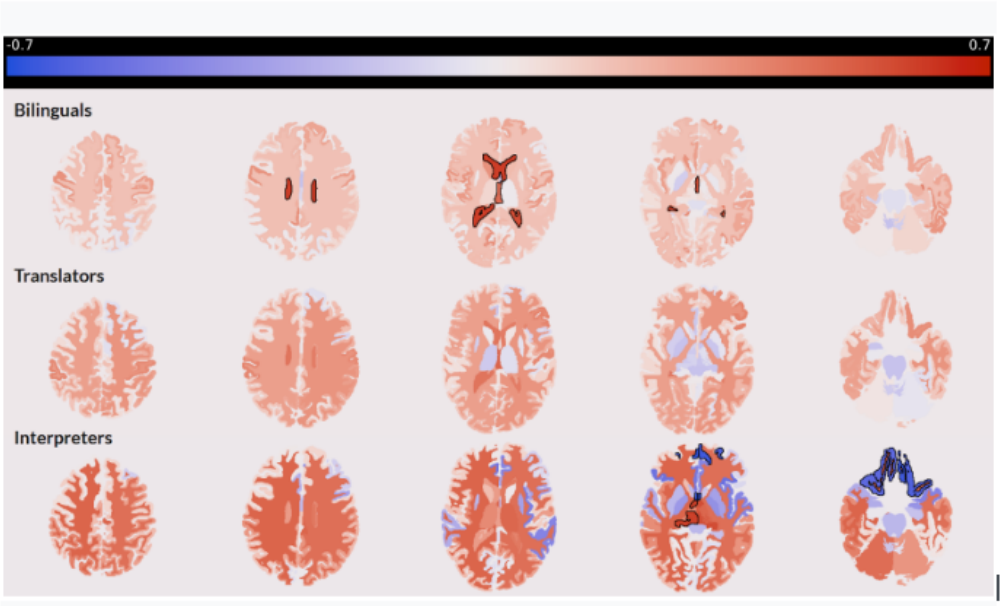

Figure 2: ROI-wise maps from the BrainAge model. Partial Pearson correlation between bias-corrected brain-age gap and residualized ROI volume (ICV, age and sex regressed out). Red = positive, blue = negative. ROIs outlined in black survive FDR-controlled significance.

References

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